

Patrick Semler

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EDUCATION

Texas A&M University

Bachelor of Science in Computer Science

College Station, TX

Expected May 2028

- **GPA:** 3.79/4.00 | **Honors:** National Merit Scholar; President's Endowed Scholar
- **Coursework:** Program Design and Concepts; Data Structures and Algorithms; Computer Organization; Programming Languages; Discrete Math; Linear Algebra
- **Organizations:** Aggies Create; Artificial Intelligence for Sustainable Food Systems (AISFS); Aggie Data Science Club; Product@TAMU; TAMUhack; TIDAL

EXPERIENCE

Research Software Engineer

Texas A&M University (AISFS)

Aug. 2026 – Present

College Station, TX

- Built a five-stage U.S. beef-supply-chain simulator with weighted representative agents, computing a scenario of 30 million head over 10 years and 250 trials in under 0.6 seconds
- Automated a pipeline that refreshes data from four USDA ERS sources, processing 179,206 rows into 11 complete annual profiles

Machine Learning Research Engineer

Texas A&M University (AISFS)

June 2026 – Aug. 2026

College Station, TX

- Cut a random-forest model's peak memory 74% (2.6 GB → 689 MB) and its size 4×, enabling Vercel deployment, while preserving accuracy (90.83% vs. 91.06%) by tuning tree count and leaf constraints
- Engineered a veterinary RAG pipeline with Voyage embeddings, Supabase pgvector, and dual-query reciprocal-rank fusion, boosting correct-source retrieval nearly 1.5× over single-query search
- Expanded diagnostic coverage to swine and equine with 64 condition mappings, bringing the inference pipeline to 8 animal categories

Software Engineer Intern

Finch

June 2026 – Aug. 2026

College Station, TX

- Rebuilt Finch onboarding as a resume-first flow and launched a new job board, cutting signup-to-dashboard time ~99% in automated testing
- Built a job-matching pipeline for hundreds of users, ranking jobs by fit, recency, and network signals with tailored resumes and cover letters
- Fixed 4 reliability failures in the application pipeline with progress persistence, write retries, duplicate prevention, and reliable worker cancellation

PROJECTS

LeetMind | TypeScript, Python, React 19, FastAPI, PostgreSQL, Docker

2026

- Built an adaptive coding platform that tracks Elo ratings across 20 algorithm concepts and generates problems targeted to each learner's skill gaps
- Validated AI-generated problems by having a separate model independently solve each one, publishing only those where both solutions matched on all authored tests and 50 randomized inputs

LiftLedger | TypeScript, React 19, Hono, PostgreSQL, Vercel AI SDK, Vercel

2026

- Built and deployed a fitness dashboard combining Hevy workouts, Telegram meal logging, and Apple Health, with a muscle-heatmap body chart, goal tracking, and a tool-calling AI assistant
- Designed a Telegram food logger that turns meal photos and text into calorie and macro entries using an LLM, grounding values in USDA data

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, SQL

Web and APIs: React, Vite, Tailwind CSS, FastAPI, Flask, Django, Hono, Node.js, REST APIs

Data and Infrastructure: PostgreSQL, Supabase, Docker, Cloudflare Workers, Vercel

Machine Learning: scikit-learn, ONNX Runtime

AI Engineering: LLM Application Development, Retrieval-Augmented Generation (RAG), Vercel AI SDK

Testing and Tools: Git, GitHub, AI Coding Assistants (Claude Code, Codex), pytest, Vitest, Playwright, Testing Library